ERIC TRINQUE

Software Engineer
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PROFESSIONAL SUMMARY

Software engineer specializing in concurrent systems, algorithm design, and full-stack development with expertise in **Go**, **C++**, **Kotlin**, and **JavaScript/TypeScript**. Proven ability to architect **scalable solutions**, **optimize performance**, and deliver **production-ready** applications. **4+ years** managing complex projects with **cross-functional** teams and maintaining **90%+** client satisfaction rates.

TECHNICAL SKILLS

Programming Languages: Go, C++, Kotlin, JavaScript, TypeScript, Python, SQL
Frameworks & Libraries: React, Node.js, wxWidgets, JavaFX, Bootstrap, TailwindCSS
Tools & Platforms: Docker, Kubernetes, Git, GitHub, PostgreSQL, MySQL, Linux, REST APIs
Practices & Methodologies: Agile/Scrum, CI/CD, Test-Driven Development, Code Review, Unit Testing, Integration Testing, Version Control, Debugging, Microservices Architecture, Object-Oriented Design, Algorithm Optimization

PROFESSIONAL EXPERIENCE

Software Developer (Freelance) | Remote, FL | 2024 - Present

- **Engineered enterprise intranet platform serving 50+ concurrent users** with **99.5%** uptime using Go REST API backend, React/TypeScript frontend, and PostgreSQL database, reducing manual reporting time by **40%**
- **Architected secure authentication system** implementing JWT token-based authorization and role-based access control (RBAC), eliminating unauthorized access incidents across 3 admin tiers
- **Automated deployment pipeline** using Git version control and CI/CD workflows, reducing deployment time from 2 hours to **15 minutes** while maintaining zero-downtime releases
- **Collaborated in Agile sprints** with stakeholders to gather requirements, conduct code reviews, and deliver **12+** feature releases based on user feedback, achieving **95%** feature adoption rate

AI Training Specialist - Software Engineering | Alignerr AI & Outlier AI | Remote, FL | 2024 - Present

- **Conducted 500+ code reviews** of AI-generated solutions across **Go**, **C++**, **Python**, **TypeScript**, and **JavaScript**, achieving **95%** accuracy in identifying correctness issues, security vulnerabilities, and performance bottlenecks
- **Designed and executed comprehensive test suites** including unit tests, integration tests, and edge case validation across 8 programming paradigms, improving model output quality by **30%**
- **Utilized advanced debugging tools and profilers** (GDB, Chrome DevTools) to identify memory leaks, race conditions, and performance issues in generated code, providing detailed technical feedback for model training
- **Collaborated with distributed engineering teams** across 4 time zones using Agile methodologies, meeting **98%** of sprint commitments while maintaining strict quality benchmarks

Owner/Operator | Resolvd LLC | Gainesville, FL | 2020 - 2024

- **Managed end-to-end project delivery for 200+ clients** with aggregated project value exceeding **$15M**, maintaining **100%** on-time delivery rate and **97%** budget adherence through systematic scope management and risk mitigation
- **Led cross-functional teams of 5-15 contractors and vendors**, establishing standardized communication protocols and project tracking systems that reduced issue resolution time by **60%**
- **Utilized management dashboards and reporting systems** to track KPIs, resource allocation, and milestone progress, leading to **25%** improvement in productivity
- **Negotiated and managed vendor relationships and supply chain logistics**, optimizing resource allocation and reducing project costs by **18%** while maintaining quality standards

Real Estate Agent (Independent) | Pepine Realty | Gainesville, FL | 2021 - 2024

- **Maintained 92% client retention rate** across **50+** transactions by implementing CRM-based follow-up systems and data-driven market analysis, resulting in **60%** referral-based business growth
- **Managed complex negotiations** in high-pressure environments involving multiple stakeholders, contract terms, and financial constraints, closing **50+** transactions with average **98%** list-to-sale price ratio

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TECHNICAL PROJECTS

Go-Crawl: Concurrent Web Crawler | Go-Crawl
Go, Goroutines, Concurrency, HTML Parsing, Worker Pool Pattern

- **Built production-grade web crawler processing 10,000+ pages** using goroutine-based worker pool architecture with configurable concurrency (**1-100** workers), achieving **200** pages/second throughput while maintaining thread-safety via mutex synchronization
- **Engineered robust HTML parsing pipeline** with comprehensive error handling, retry logic, and custom merge sort algorithm ($O(n \log n)$ complexity) for link frequency analysis, achieving **85%** test coverage through unit and integration testing
- **Implemented concurrent-safe data structures** utilizing Go's sync package (Mutex, WaitGroup) to prevent race conditions, validated through race detector and stress testing under high load conditions

Boids Flocking Simulation: Real-Time Physics Engine | Boids
C++17, wxWidgets, Object-Oriented Design, Graphics Programming

- **Developed interactive particle system simulating emergent behavior** using Reynolds' flocking algorithm with 3 behavioral rules (alignment, cohesion, separation), maintaining consistent **60+ FPS** performance with **500+** concurrent entities
- **Optimized collision detection using spatial partitioning**, reducing computational complexity from $O(n^2)$ to $O(n \log n)$, enabling real-time parameter adjustment with **zero frame drops**
- **Engineered extensible OOP architecture** following **SOLID** principles with custom 2D vector mathematics library, file I/O serialization system, and GUI controls
- **Implemented state persistence system** with binary serialization enabling save/load of simulation configurations, supporting import of custom initial conditions and export of behavioral patterns

Fractal Generator: Mathematical Visualization Platform | Frac-Gen
Kotlin, JavaFX, Recursive Algorithms, Graphics Rendering

- **Created interactive desktop application** rendering 5+ fractal types (Sierpinski Triangle/Carpet, Mandelbrot Set, Julia Set) with configurable parameters including recursion depth (1-25 iterations), rotation angles, and RGB color schemes
- **Optimized recursive rendering algorithms**, achieving 30% performance improvement through memoization and tail-call optimization, enabling smooth real-time visualization at 30+ FPS for deep recursion levels
- **Designed modular MVC architecture** with clean separation of concerns between rendering engine, UI controls, and fractal algorithm implementations, enabling addition of new fractals with <50 lines of code
- **Implemented responsive UI** using JavaFX scene graph with real-time controls (sliders, color pickers, spinners) and canvas rendering supporting arbitrary resolutions up to 4K

EDUCATION

Bachelor of Science in Computer Science | Full Sail University | Expected April 2027 | **GPA: 3.75**
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering, Memory Management, AI/Machine Learning

Professional Development:
Frontend Masters - Web Development Specialization (JavaScript, TypeScript, React, Node.js)
Boot.Dev - Backend Engineering & Systems Programming (Go, Python, Databases, Concurrency)